

Baseline Machine Learning from Scratch using AirQualityUCI

1. Machine Learning Problem Types

Machine learning problems are generally divided into supervised and unsupervised learning.

- **Supervised learning** uses data where the correct answers are already known. During training, the model learns the relationship between the input features and the target so it can make predictions on new, unseen data.
- **Unsupervised learning** works differently because there are no correct answers provided. Instead, the model explores the data to discover patterns, similarities, or groups on its own.

Within these categories, there are three common problem types:

- **Regression** predicts a continuous numerical value, such as house prices, temperature, or CO(GT) concentration.
- **Classification** predicts one category from a fixed set of possible classes, such as spam/not spam or high/low pollution.
- **Clustering** groups similar data points together without using labels, helping identify natural patterns in the dataset.

2. Dataset Framing

A machine learning dataset is made up of several important components:

- A **sample** is a single observation or row in the dataset.
- A **feature** is one of the input variables the model uses to make predictions.
- A **label** is the correct class assigned to a sample in a classification task.
- The **target** is the value the model is trying to predict. For regression, the target is a numerical value, while for classification it is a class label.
- All input features are stored in the **feature matrix (X)**, where each row represents one sample and each column represents one feature.
- The **target vector (y)** stores the corresponding target value or label for every sample

3. Regression

Regression is used whenever the goal is to predict a numerical value. After making a prediction, the difference between the actual value and the predicted value is called the **residual**. Smaller residuals generally indicate better predictions.

To measure how well a model is performing, we use a **loss function**, which calculates how far the predictions are from the true values. During training, the model tries to reduce this loss as much as possible.

A simple **baseline regression model** predicts the average target value for every sample. Any useful regression model should perform better than this simple baseline.

One of the most common regression algorithms is **linear regression**, which fits the best possible straight line through the data to capture the relationship between the features and the target.

4. Classification

Classification is used when the goal is to predict categories instead of numbers.

- In **binary classification**, there are only two possible outcomes, such as yes/no or positive/negative.
- **Logistic regression** is a commonly used classification algorithm. Instead of predicting a number directly, it estimates the probability that a sample belongs to a particular class.

A **decision threshold**, usually set to 0.5, converts this probability into the final predicted class.

For comparison, a **majority-class baseline** simply predicts the most common class for every sample.

5. Clustering

Clustering is an unsupervised learning technique that groups similar data points together. The **KMeans** algorithm begins by placing several cluster centers, called **centroids**, in the data. Each sample is assigned to the nearest centroid. The centroids are then updated based on the assigned samples, and this process repeats until the cluster centers stop changing significantly. Once clustering is complete, the clusters are interpreted by examining the characteristics shared by the samples within each group.

6. Gradient Descent

Most machine learning models learn by gradually improving their parameters during training. **Gradient descent** is an optimization algorithm that repeatedly adjusts the model's parameters, such as weights and bias, to reduce the loss function. The **learning rate** determines how large each update should be. If the learning rate is too high, the model may overshoot the best solution. If it is too low, learning becomes very slow. This process continues over many **iterations** until the model **converges**, meaning further updates produce little improvement. Feature scaling is also important because features with very different numerical ranges can make gradient descent slower or less stable.

7. Train, Validation and Test Sets

To build a reliable model, the dataset is usually divided into three parts.

- The **training set** is used to teach the model.

- The **validation set** helps choose the best settings, such as the learning rate or other hyperparameters.
- The **test set** is kept completely separate and is only used once, after training is finished, to estimate how well the model performs on new data.

Using the test set repeatedly while making improvements can lead to overly optimistic results because the model gradually becomes tuned to that specific test data.

8. Evaluation Metrics for Regression

Different metrics are used to evaluate regression models.

- **MAE (Mean Absolute Error)** measures the average size of the prediction errors without giving extra importance to larger mistakes.
- **MSE (Mean Squared Error)** squares each error before averaging, so larger errors have a much greater impact.
- **RMSE (Root Mean Squared Error)** is simply the square root of MSE, making the error easier to interpret because it has the same units as the target.
- **R-squared (R^2)** measures how much of the variation in the target variable is explained by the model. Higher values generally indicate a better fit.

9. Evaluation Metrics for Classification

Several metrics are commonly used to evaluate classification models.

- **Accuracy** measures the percentage of predictions that are correct overall.
- **Precision** measures how many predicted positive samples are actually positive.
- **Recall** measures how many actual positive samples the model successfully identifies.
- **F1-score** combines precision and recall into a single metric, making it useful when both are equally important.
- A **confusion matrix** provides a detailed breakdown of correct and incorrect predictions by showing true positives, true negatives, false positives, and false negatives.

10. Evaluation Metrics for Clustering

Since clustering has no correct labels, different evaluation methods are used.

- **Inertia** measures how close the data points are to their assigned centroids. Lower values usually indicate more compact clusters.
- **Silhouette score** measures how well each sample fits within its own cluster compared to nearby clusters. Scores closer to 1 indicate better separation between clusters.
- **Cluster counts** show how many samples belong to each cluster, helping identify clusters that may be unusually small or large.
- Finally, **visual inspection** of the clustered data can provide a simple way to judge whether the clusters appear meaningful and well separated.

11. Baseline Predictors

A baseline predictor is the simplest possible model that can be built. For regression, this might mean always predicting the average target value. For classification, it could mean always predicting the majority class. Comparing a trained model against a baseline helps determine whether it has actually learned useful information from the data.

12. Data Leakage

Data leakage happens when information that would not normally be available during prediction is accidentally used while training the model. This can make the model appear much more accurate than it really is. Common examples include using features derived from the target value, applying preprocessing to the entire dataset before splitting it, repeatedly tuning the model using the test set, or including columns that were created from the correct labels.

13. Limitations of Baseline Machine Learning

Performance may suffer if the dataset is too small, the features contain a lot of noise, the target values are incorrect, or the available features do not contain enough useful information. Other challenges include highly imbalanced datasets and relying on evaluation metrics that do not properly reflect the model's performance.

Questions and Answers

1. What is the difference between regression, classification, and clustering?

Regression predicts a numerical value that can take many possible values, such as CO(GT) concentration. Classification predicts one category from a fixed set of classes, such as "high" or "low." Clustering does not use labels at all. Instead, it groups similar data points together based on their characteristics.

2. What are features and targets?

Features are the input variables that the model uses to make predictions. The target is the value the model is trying to predict, such as the actual CO(GT) concentration or a class label.

3. What is the difference between supervised and unsupervised learning?

Supervised learning uses training data with known target values, allowing the model to learn from correct answers. Unsupervised learning has no target labels, so the model looks for patterns or relationships within the data on its own.

4. What is a baseline predictor?

A baseline predictor is the simplest possible prediction method. Examples include always predicting the average value in regression or always predicting the majority class in classification. It provides a minimum level of performance that more advanced models should improve upon.

5. Why is a model weak if it does not beat the baseline?

If a trained model cannot outperform a simple baseline, it suggests that the model has not learned useful relationships from the features. In that case, the additional complexity does not provide any real benefit.

6. Why do we need train, validation, and test sets?

The training set is used to teach the model, the validation set is used to improve and tune it, and the test set provides a final evaluation using completely unseen data. Separating the data this way helps measure how well the model is likely to perform in real-world situations.

7. Why should the test set not be used repeatedly during model selection?

Repeatedly checking the test set while making changes to the model gradually causes the model to fit that specific test data. This produces overly optimistic results that may not reflect performance on genuinely new data.

8. What is gradient descent and what role does the learning rate play?

Gradient descent is an optimization algorithm that updates a model's parameters to reduce the loss function. The learning rate controls how large each update is during training.

9. What happens if the learning rate is too high or too low?

If the learning rate is too high, the model may overshoot the best solution and training can become unstable. If it is too low, training becomes much slower and may take a long time to reach a good solution.

10. What is the difference between MAE, RMSE, and R-squared?

MAE measures the average prediction error and treats all errors equally. RMSE also measures prediction error but gives greater importance to larger mistakes. R-squared measures how much of the variation in the target variable is explained by the model.

11. What is the difference between accuracy, precision, recall, and F1-score?

Accuracy measures overall correctness. Precision measures how many predicted positive cases are actually positive. Recall measures how many actual positive cases the model successfully identifies. F1-score combines precision and recall into a single value that balances both metrics.

12. Why is accuracy misleading for imbalanced classification?

When one class is much more common than the other, a model can achieve high accuracy simply by always predicting the majority class. Even with high accuracy, it may completely fail to detect the minority class, making it unreliable.

13. What does a confusion matrix show?

A confusion matrix shows the number of true positives, true negatives, false positives, and false negatives. It helps identify the specific types of mistakes a classification model is making.

14. What is inertia in KMeans?

Inertia is the total squared distance between each data point and the centroid of the cluster it belongs to. Lower inertia generally indicates that the clusters are more compact.

15. What is silhouette score?

The silhouette score measures how well each data point fits within its assigned cluster compared to other nearby clusters. Values closer to 1 indicate better-defined and better-separated clusters.

16. What is data leakage? Give three examples relevant to this task.

Data leakage occurs when information that would not normally be available during prediction is accidentally used during training, leading to unrealistically high performance.

Examples include:

- Using a feature that is derived directly from the target.
- Scaling the entire dataset before splitting it into training and testing sets.
- Repeatedly tuning the model based on the test set.